

Response Geometry in Integrated Gradients: A Diffusion-Policy Case Study

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Revision status (October 1, 2026). This working draft¹ reports analytic results, a canonical retrospective reanalysis and an incomplete numerical diagnostic campaign. Comparative ranking efficacy and learned-weight dependence are outside its empirical claims. Final evidence integration and independent review remain open; it is not submission-ready.

Abstract

Attribution evaluation depends on both the ranking that selects interventions and the response that scores them. We analyze this distinction for Integrated Gradients (IG) of a fixed-noise diffusion policy’s predicted action chunk. For quadratic and stabilized negative- ℓ_2 action-discrepancy targets, we derive an exact normalized-response relation and the positive, path-dependent factor relating their gradients. A smooth two-feature construction reverses the IG ranking despite pointwise collinearity of the nonzero gradients, including at the implemented stabilizer. An equal-feature construction gives continuous quadratic-response insertion and deletion AUC of 2/3 for every ordering, ruling out a universal random-ranking reference of 0.5. A canonical reanalysis of 99 saved Robotics Diffusion Transformer files then separates fixed-ranking response changes from unauthenticated cross-target comparisons. For the same recorded quadratic-target rankings, changing only the response changes median deletion AUC from 0.4479 to 0.2908 without changing a selected feature. Conversely, quadratic rescaling changes the recorded ℓ_2 rankings’ deletion AUC from 0.2830 to 0.4404. Existing actual-model diagnostics additionally show repeatable computations with unresolved integration-budget sensitivity in an explicitly incomplete cohort. They do not qualify a production numerical setting. Together these results distinguish response geometry, numerical reliability and ranking quality. They establish neither improved attribution rankings nor explanations of task success or learned-weight dependence. The artifact links reconstructed results to preserved records, retains provenance limitations and numerical failures, and provides the analytic checks.

1 Introduction

Vision-language-action (VLA) models such as RT-2 (Zitkovich et al., 2023), Octo (Octo Model Team et al., 2024), and the Robotics Diffusion Transformer (RDT) (Liu et al., 2025) combine sensory observations and instructions to predict robot actions. Diffusion policies (Chi et al., 2023) generate action chunks through iterative denoising (Ho et al., 2020). Applying a scalar attribution method to this vector-valued output requires specifying what aspect of the action prediction is being explained. That choice also matters when evaluating the resulting explanation.

Integrated Gradients (IG) (Sundararajan et al., 2017) integrates a scalar output’s gradients from a baseline to an input. For a fixed-noise action sample, one possible scalar is its discrepancy from a detached reference action. A quadratic discrepancy and a negative norm express related notions of change, but they define different scalar functions. A perturbation evaluation introduces a second choice: the response used to score

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the feature removals or insertions selected by an attribution ranking. A comparison that changes both choices cannot identify a ranking improvement from its native-response AUC alone.

We examine this problem through exact derivations, constructive examples and a retrospective RDT case study. Changing the response from a quadratic discrepancy to a stabilized negative ℓ_2 discrepancy changes normalized AUC even when the selected features and perturbed outputs stay fixed. At the same time, the targets’ positive, path-dependent gradient relationship permits their IG rankings to differ. These statements address distinct operations: rescoring existing interventions and recomputing attributions. Neither establishes which ranking is more useful under a common response.

A third issue is whether a computed attribution accurately approximates the specified mathematical quantity. Small scalar completeness residuals can conceal coordinate errors, and repeated finite-precision calculations can reproduce the same inaccurate approximation. We report already collected numerical diagnostics with their full planned roster and unfinished cells, rather than using a subset of successful checks to certify the implementation. The intended attribution is a fixed-context action-discrepancy quantity; it does not explain eventual task success or establish safe robot behavior.

Our contributions are as follows.

1. **Response and path-weight analysis.** An exact normalized-response identity, its overshoot and zero-gap qualifications, and a finite-stabilizer construction in which the quadratic and negative-norm targets reverse IG feature order.
2. **A calibration counterexample.** An equal-contribution policy for which every ordering has continuous quadratic-response insertion and deletion AUC of $2/3$, so 0.5 is not a universal chance reference.
3. **An authenticated retrospective case study.** Within-ranking rescoring of preserved RDT perturbation curves, with explicit occurrence rules, episode grouping and result-to-record lineage. Unknown historical context identities prevent a paired comparison of ranking quality.
4. **Numerical diagnostic evidence.** Fixed-input repeatability and integration-budget checks from a partially completed campaign, with failures and incomplete coverage retained. These diagnose limitations of the tested calculations without approving a production setting.

The contribution is the concrete separation of these issues and its documented consequences in one diffusion-policy family. A common-response ranking study would answer an additional efficacy question; its absence limits that conclusion without invalidating the fixed-ranking analysis.

2 Related Work

2.1 Attribution Methods

Sundararajan et al. (2017) introduced IG with the sensitivity and implementation-invariance axioms and showed that straight-line path integration satisfies both. Later work has addressed limitations of the vanilla method. Jha et al. (2021) computed attributions over neural stochastic differential equation models to produce smoother attribution maps, and later showed that shaping the injected noise further improves robustness (Jha et al., 2022). Walker et al. (2024) observed that gradients from saturation regions contribute noise and proposed Integrated Decision Gradients, which concentrates integration on the decision-relevant portion of the path. For vision transformers, metric-driven attributions (Walker et al., 2025b) and information-flow decompositions (Walker et al., 2025a) improve patch-level explanations. Guided IG (Kapishnikov et al., 2021) adapts the integration path to reduce pixel-level noise. Petsiuk et al. (2018) introduced the insertion and deletion curves we adopt as ranking-based faithfulness metrics. The reliability of perturbation metrics is itself an active concern. Hooker et al. (2019) evaluate attributions by retraining on ablated inputs. Rong et al. (2022) show that deletion without retraining leaks information through the shape of the mask itself, and Tomsett et al. (2020) find that common saliency metrics are themselves unreliable. Our case study makes

this distinction explicit for action-discrepancy responses: it relates their normalized curves algebraically and rescores preserved interventions while holding their feature ordering fixed. Metric dependence itself is already established by this literature. Adebayo et al. (2018) proposed model and data randomization tests to assess whether saliency maps depend on learned parameters and training data. Their results caution against interpreting visually plausible maps as evidence of that dependence. Our backbone randomizations are limited tests because learned encoders and adaptors remain in place. For diffusion generative models in the image domain, cross-attention maps relate generated pixels to prompt tokens (Tang et al., 2023). This differs from attributing a scalar discrepancy of a predicted control chunk.

Our implementation applies IG to a fixed-noise action-discrepancy function with modality-specific baselines. Path weighting is also an established attribution idea, as in IDG (Walker et al., 2024); the factor derived here follows specifically from changing the scalar discrepancy target. The analysis below specializes these established concerns to quadratic and stabilized negative-norm action discrepancies, tracing their separate effects on fixed interventions and integrated rankings.

2.2 Vision-Language-Action Models

VLA models unify perception, language understanding, and action generation. RT-2 (Zitkovich et al., 2023) fine-tunes a vision-language model to emit discretized actions as token sequences. Octo (Octo Model Team et al., 2024) trains a generalist policy on a large multi-robot dataset with a transformer backbone. Diffusion Policy (Chi et al., 2023) frames action generation as conditional denoising diffusion (Ho et al., 2020). It produces action chunks by iterative refinement rather than single-step regression. RDT (Liu et al., 2025) scales this approach with a Diffusion Transformer (DiT) backbone, a SigLIP vision encoder (Zhai et al., 2023), and a T5-XXL language encoder (Raffel et al., 2020). RDT processes six camera slots (three camera views over a two-frame history) through SigLIP into 4,374 vision tokens. It combines them with up to 1,024 language tokens and a 128-dimensional unified state vector, and it generates 64-step action chunks through a five-step DPM-Solver++ (Lu et al., 2025) denoising schedule. This multimodal, multi-step architecture makes RDT a compelling but challenging target for attribution.

2.3 Interpretability in Robotics

Prior work on explaining robotic policies has focused on reinforcement learning agents in discrete-action settings. Greydanus et al. (2018) applied perturbation-based saliency to Atari agents, and counterfactual methods (Amitai et al., 2024) contrast the outcomes of executed and alternative actions for policy explanation. In continuous control, VisualBackProp (Bojarski et al., 2018) visualizes convolutional neural network (CNN) saliency for end-to-end driving policies. That work addresses convolutional driving models rather than a multimodal diffusion sampler. Recent VLA studies examine different explanation targets. Zhang et al. (2026) estimate visual influence using randomized perturbations and action-discrepancy responses, with explicit assumptions for relating mean-squared error to distributional divergence. VLA-Trace (Shi et al., 2026) combines representation comparisons, attention interventions and behavioral probes. IMPACT-VLA (Kim & Park, 2026) attributes phase-modality interventions to downstream task success through counterfactual closed-loop execution. Our estimand is the discrepancy of a fixed-noise predicted action chunk at a fixed context, not an explanation of eventual task success. The present study examines how response geometry affects the evaluation of IG rankings for this estimand; it does not claim to originate VLA interpretability.

3 Attribution Targets and Evaluation

3.1 Integrated Gradients Preliminaries

Given a differentiable function $F: \mathbb{R}^n \rightarrow \mathbb{R}$, an input $x \in \mathbb{R}^n$, and a baseline $x' \in \mathbb{R}^n$, the integrated gradient along dimension i is

$$\text{IG}_i(x, x') = (x_i - x'_i) \int_0^1 \frac{\partial F}{\partial x_i} \Big|_{x' + \alpha(x - x')} d\alpha. \quad (1)$$

Under the fundamental-theorem and chain-rule assumptions along the path, IG satisfies *completeness*: $\sum_i \text{IG}_i(x, x') = F(x) - F(x')$. We approximate the integral with a uniform Riemann sum of m steps, giving

$$\text{IG}_i \approx \frac{x_i - x'_i}{m+1} \sum_{r=0}^m \frac{\partial F}{\partial x_i} \Big|_{x' + \frac{r}{m}(x-x')}. \quad (2)$$

This equal-weight endpoint average describes the historical implementation. Convergence requires regularity of the path integrand; finite-precision accumulation introduces additional error that increasing m does not isolate or necessarily reduce.

3.2 A Deterministic Target Through the Denoising Chain

At policy call t , write the sampler output as $g_\theta(o_t, z_t)$ for observation o_t and fixed diffusion noise z_t . We denote this predicted action chunk by μ_t for consistency with historical notation; it is a sample conditioned on fixed noise, not an identified conditional mean. The reference chunk $a_t = g_\theta(o_t, z_t)$ is computed once without gradients. For an intervened input, the primary attribution target is the auxiliary score

$$Q_t = -\frac{1}{2\sigma^2 D} \|a_t - \mu_t\|^2, \quad (3)$$

with $\sigma^2 = 1$ and $D = 512$ active entries: 64 predicted time steps over eight active dimensions. This is the nonconstant part of a Gaussian log kernel averaged over entries, with its log-normalizing constant omitted. It has not been derived as the diffusion policy’s conditional log likelihood. Historical code and files call this target `logpi`; we use ΔQ for its change. Attribution concerns the complete predicted chunk, of which the execution loop applies a subsampled prefix before replanning or termination.

We hold the noise realization fixed across the interpolation path to evaluate one deterministic scalar function. Attribution of an expectation over noise would be a different estimand and requires a justified gradient-expectation interchange and Monte Carlo treatment. Independent noise at each path point does not compute the fixed-noise quantity considered here. The historical implementation replaced nonfinite gradient entries by zero. That operation can corrupt individual coordinates while cancellation leaves the scalar completeness residual small or even zero; completeness is not a certificate of coordinate accuracy.

3.3 Alternative Scalar Targets

We compare the quadratic target with two alternative action-discrepancy targets,

$$y_t^{\ell_2} = -\sqrt{\|a_t - \mu_t\|_2^2 + \epsilon}, \quad y_t^{\max} = -\max_j |a_{t,j} - \mu_{t,j}|, \quad (4)$$

where $\epsilon = 10^{-12}$ is the implemented stabilizer and both targets use the same D active entries. The stabilized ℓ_2 target is smooth in the residual and has zero gradient at exact agreement; away from the stabilizer scale its action-space gradient approaches unit norm. The maximum-deviation target, abbreviated `maxdev`, uses a different norm and is not a monotone transform of ℓ_2 in general: residuals (1, 1) and (1.1, 0) reverse order. The quadratic and stabilized ℓ_2 responses are algebraically related for a fixed perturbation, although their gradients can induce different attribution rankings. Their units and scales differ, so a larger magnitude alone does not demonstrate greater information or operational usefulness. Section 5.1 isolates response changes for the quadratic and stabilized ℓ_2 targets. The maximum-deviation records remain an auxiliary diagnostic in the released analysis.

3.4 Response Transforms and Attribution Path Weights

Fix the reference action, active coordinates, model, noise and other modalities, and write $r(x) = \|a - \mu(x)\|_2^2$ and $L(x) = -\sqrt{r(x) + \epsilon}$. Where the gradients exist, the chain rule gives the exact identity

$$\nabla Q(x) = \frac{\sqrt{r(x) + \epsilon}}{D\sigma^2} \nabla L(x). \quad (5)$$

Thus nonzero pointwise gradients have the same direction. At exact self-reference both vanish. Along $x(\alpha) = x' + \alpha(x - x')$, however,

$$\text{IG}_i^Q = (x_i - x'_i) \int_0^1 \frac{\sqrt{r(x(\alpha)) + \epsilon}}{D\sigma^2} \partial_i L(x(\alpha)) d\alpha. \quad (6)$$

The positive weight generally varies with α . When coordinate contributions vary differently along the path, their integrated rankings can therefore change. This does not contradict IG implementation invariance, which compares implementations of the same scalar function (Sundararajan et al., 2017).

A concrete example uses the scalar action $\mu(x_1, x_2) = x_1^2 + \frac{4}{5}x_2$, baseline $(0, 0)$, actual input $(1, 1)$, detached reference $a = \frac{9}{5}$, and $D = \sigma^2 = 1$. On the same straight path, exact quadratic-target integration yields

$$\text{IG}^Q = \left(\frac{23}{30}, \frac{64}{75} \right), \quad \lim_{\epsilon \downarrow 0} \text{IG}^L = \left(1, \frac{4}{5} \right). \quad (7)$$

Quadratic IG ranks feature 2 first, whereas stabilized L with the implemented $\epsilon = 10^{-12}$ ranks feature 1 first with a margin at least 0.1999975. Appendix A proves the finite-stabilizer bound, so the reversal does not rely on replacing the implemented response by its limit. This establishes existence in a smooth nonlinear policy, not a universal rank change or an empirical RDT effect.

3.5 Per-Step Episode Attribution

RDT outputs action chunks of 64 steps, subsampled $4\times$ to 16 environment steps. We run IG once per *policy call*, not per environment step, because the chunk is the atomic unit of decision. The attribution unit is therefore one policy query. Runtime depends on the implementation, hardware, precision and integration budget; the earlier timing summaries are not used as a performance result.

At each policy call we compute three independent attributions through modality-specific forward functions.

- **Vision.** Freeze language and state and interpolate the adapted image tokens from the baseline to the real observation.
- **Language.** Freeze vision and state and interpolate the adapted language tokens from a minimal two-token baseline encoded through T5-XXL to the real instruction embedding.
- **State.** Freeze vision and language and interpolate the 128-dimensional unified state vector (eight active dimensions for the Franka arm, namely seven joints plus the gripper) from zeros to its real value. The state-adaptor multilayer perceptron (MLP) sits inside the forward function to preserve per-joint structure.

Algorithm 1 summarizes the procedure.

Baseline selection. The vision baseline is a uniform gray image in the external-camera slot only. The remaining five camera slots carry constant background images and are frozen at their real values, so IG attributes only to the informative slot. The language baseline is a minimal two-token sequence (the tokenizer’s padding and end-of-sequence tokens, since T5 has no beginning-of-sequence token) encoded through T5-XXL rather than a zero vector. The encoder output is zero-padded to the stored embedding length, and the language forward passes attend over the task’s real token span. The attended baseline is therefore the encoded token pair followed by zero vectors at the remaining attended positions. A zero vector is numerically permitted by the GELU-based language adaptor but defines a different representation baseline; preliminary efficacy comparisons are not independently reconstructible. The state baseline is a zero vector in the normalized joint space; its physical support and usefulness as an explanation reference are not established by this choice. Appendix B.2 reports the available vision-baseline sensitivity.

Algorithm 1 Fixed-noise per-policy-call IG construction. Historical artifact availability is qualified in Section 4.

Input. Policy π_θ , environment $\mathcal{E}$, integration steps m , target $y \in \{Q, \ell_2, \text{maxdev}\}$

Output. Attributions $\{A_t^V, A_t^L, A_t^S\}$ at every policy call.

```

1:  $o_0 \leftarrow \mathcal{E}.\text{reset}()$ 
2: for policy call  $t = 0, 1, \dots$  do
3:   Encode  $o_t$  into vision  $v_t$ , language  $l_t$ , state  $s_t$ 
4:   Draw and store  $z_t$ ; compute  $a_t \leftarrow g_\theta(o_t, z_t)$  without gradients
5:   Define  $F$  from  $a_t$ , the fixed-noise predicted chunk  $\mu_t$ , and the chosen target
6:   for modality  $M \in \{V, L, S\}$  do
7:     Set baseline  $x'_M$ , input  $x_M$ 
8:     Freeze the other two modalities
9:      $A_t^M \leftarrow \text{IG}(F_M, x_M, x'_M, m)$ 
10:  end for
11:  Record  $A_t^V, A_t^L, A_t^S$  and completeness errors
12:  Apply action chunk and advance  $\mathcal{E}$  by up to 16 steps, stopping on termination
13: end for
```

Eq. 2

3.6 Evaluation Protocol

Completeness. For each modality, with the other inputs frozen, we report the absolute residual $|\sum_i \text{IG}_i - (F(x) - F(x'))|$ and its relative version

$$\text{Err} = \frac{|\sum_i \text{IG}_i - (F(x) - F(x'))|}{|F(x) - F(x')|}. \quad (8)$$

An exactly zero signed gap makes the relative quantity undefined. A small nonzero gap may amplify numerical error. The historical action-norm filter $\|a_t\| \geq 15$ does not bound this denominator, and is retained only as a population sensitivity. A low residual checks a scalar accounting identity; canceling coordinate errors can still change the attribution map or ranking.

Perturbation curves. We rank eligible feature groups by the sum of absolute coordinate attributions. The groups are external-camera patch tokens for vision and attended instruction tokens for language. Deletion replaces a prefix of the ranking with baseline features; insertion replaces a prefix of a baseline input with original features. For a scalar response R , the normalized ordinate is

$$\tilde{R}(x_k) = \frac{R(x_k) - R(x')}{R(x) - R(x')}. \quad (9)$$

We integrate on each record’s stored fraction grid. Neither the curve nor its AUC is necessarily bounded by $[0, 1]$. The historical grid requests $\{0, 1, 5, 10, 20, 30, 50, 75, 100\}\%$ of groups; rounding changes realized language-token fractions, so the saved curves cannot be treated as uniformly sampled continuous functions. Top- k score changes are $R(x_k) - R(x)$, with units defined by R . We distinguish the attribution target that generates a ranking from the response that evaluates its interventions. The earlier study-specific thresholds are not used as efficacy tests in the revised tables.

Randomization and baseline diagnostics. C1 reinitializes model parameters and correlates the resulting attributions with the originals. The standard variant changes the final action-output projection and recomputes its reference action; frozen-reference and backbone-cascade variants retain the original reference. The historical cascade simultaneously reinitializes all backbone parameter matrices of rank at least two; it retains one-dimensional parameters, encoders and adaptors. It is not a progressive layerwise cascade. C2 shuffles both external-camera pixels and language. The historical implementation uses a pre-T5 shuffled-language cache when available and otherwise permutes post-T5 embeddings; the saved records do not authenticate which mode each call used. These are different dependence tests, and C2 is not the label-randomization

experiment of Adebayo et al. (2018). Historical perturbed maps use $m = 16$ and original maps nominally use $m = 64$, so C1 also changes the numerical approximation. A baseline sensitivity compares black, gray and blurred observation features.

4 Retrospective Data and Statistical Protocol

Scope and traceability. The revised analysis authenticates 99 preserved files containing 22,966 valid records and 1 specifically identified corrupt line. Each valid physical record has a source path, original line number and byte hash. The raw inventory contains 129 repeated-key groups, of which 58 differ in scientific fields rather than only timing metadata. The last occurrence is a declared retrospective selection convention, not evidence that a conflicting attempt is correct. A separate sensitivity removes every key with a conflicting payload. The records, occurrence decisions, population membership, result identifiers and generated manuscript cells are linked in the released lineage registry. This authenticates preserved data bytes, not missing checkpoints, observations, action tensors or noise.

Campaigns and models. The principal reconstructed RDT-170M tables use the saved Month 7 verification campaign at $m = 64$, covering PickCube, StackCube, PegInsertionSide and PickSingleYCB in Man-iSkill3 (Tao et al., 2024). These are separate experiments from the unavailable original main-pass inputs. The model was pretrained RDT-170M; the saved RDT-1B PickCube campaign used a task-finetuned checkpoint. The model comparison changes training, trajectories and historical preprocessing as well as parameter count, and is not a scale or competence intervention. Appendix B therefore reports the 1B cohort separately. No missing original completeness, standard-sanity, or early PickCube/StackCube integration-budget result is copied into a primary table.

Known protocol limits. The historical action chunk is subsampled before simulator execution, while attribution concerns the full predicted chunk. The four-task scope reflects historical implementation choices. An attempted TurnFaucet run used an incorrect controller, and no corrected rerun is documented; unsuccessful outcomes under that configuration cannot justify a capability exclusion. Early follow-up jobs used a zero language baseline in place of encoded padding/end tokens, and the PickSingleYCB follow-up used a proxy PickCube instruction embedding. The pooled 1B language cohort spans the early baseline and later encoded baselines. Early sidecar filenames also omitted seed and integration budget, permitting overwrites and possible reuse across nominal runs. Seed-tagged Month 7 records mitigate that filename issue but do not recover old contexts. We do not infer exact context equality from matching scalar fingerprints.

Estimands and uncertainty. The primary population is all valid saved calls after the declared occurrence rule. Continuous call-level summaries use medians over calls; residual pass rates use proportions over the stated call population. Solver contrasts report relative changes between endpoint medians, and the separately named equal-episode analysis uses episode medians. We resample entire recorded episode groups, keyed by task, model, recorded seed and episode, with $B = 10,000$ draws and seed zero. Point estimates and intervals use exactly the same source-record population. These percentile intervals describe the released empirical episode mixture, with longer episodes contributing more calls; they do not estimate variability across independently trained checkpoints. A separately named 1B sensitivity reports the median of episode medians, weighting episodes equally. For the old randomization variants we also group potentially reused nominal task/episode keys across seeds as a conditional dependence sensitivity. Neither grouping repairs unobserved context identity. Very small cluster counts are explicitly disclosed, and two-group baseline resampling is not presented as calibrated interval coverage. All unfiltered results and the historical norm-selected sensitivity remain available; no favorable tail is removed to obtain a threshold verdict.

5 Response Geometry and Retrospective Results

5.1 Separating Attribution Target from Evaluation Response

The same recorded interventions can be rescored without recomputing IG or changing a selected feature. Let d be the recorded active-action discrepancy, $Q = -d^2/1024$, and $L = -\sqrt{d^2 + \epsilon}$ with $\epsilon = 10^{-12}$. Then

$$L = -\sqrt{-1024Q + \epsilon}, \quad Q = -\frac{L^2 - \epsilon}{1024}. \quad (10)$$

For an exact self-reference actual endpoint and an all-baseline squared residual $R > 0$, the normalized responses satisfy

$$\begin{aligned} N_Q(r) &= 1 - \frac{r}{R}, & N_L(r) &= \frac{\sqrt{R + \epsilon} - \sqrt{r + \epsilon}}{\sqrt{R + \epsilon} - \sqrt{\epsilon}}, \\ N_Q(r) - N_L(r) &= \frac{(\sqrt{r + \epsilon} - \sqrt{\epsilon})(\sqrt{R + \epsilon} - \sqrt{r + \epsilon})}{R}. \end{aligned} \quad (11)$$

Consequently $N_Q > N_L$ for $0 < r < R$, with equality at both endpoints and reversed ordering for $r > R$. Integration with nonnegative weights preserves the weak ordering if every intervention remains in range, even for a nonmonotone curve; strict area inequality requires positive integration weight on an interior residual. Overshooting points can change the area ordering and must not be clipped; if $R = 0$, both normalizations are undefined. For in-range curves, switching from Q to L therefore lowers insertion and deletion AUC together. If larger insertion AUC and smaller deletion AUC are interpreted as better ranking quality, this single response change suggests opposite quality changes for interventions that have not changed. The retrospective computation below transforms the recorded endpoints directly and does not assume exact historical self-reference. We transform every recorded curve ordinate, evaluate its endpoints under the transformed response, and reintegrate on the same saved grid. Each ranking cohort contains 750 calls from 30 episodes after duplicate handling. For quadratic-target vision rankings, median deletion AUC changes from 0.4479 under Q to 0.2908 under L . For stabilized- ℓ_2 rankings it changes from 0.2830 under L to 0.4404 under Q . Table 1 and Figure 1 include both modalities and both perturbation directions.

Ranking target	Response	Modality	Insertion AUC	Deletion AUC
Q	Q	Vision	0.883 [0.845, 0.909]	0.448 [0.408, 0.504]
Q	Q	Language	0.580 [0.510, 0.658]	0.372 [0.311, 0.428]
Q	L	Vision	0.756 [0.718, 0.785]	0.291 [0.253, 0.333]
Q	L	Language	0.469 [0.412, 0.516]	0.257 [0.228, 0.284]
L	Q	Vision	0.884 [0.833, 0.921]	0.440 [0.402, 0.501]
L	Q	Language	0.635 [0.582, 0.720]	0.368 [0.302, 0.407]
L	L	Vision	0.756 [0.700, 0.803]	0.283 [0.252, 0.327]
L	L	Language	0.507 [0.456, 0.566]	0.249 [0.202, 0.278]

Table 1: Within-ranking response transformation, with median AUC and conditional 95% episode-bootstrap intervals. Each ranking cohort contains 750 calls from 30 episodes. The same recorded interventions are reused within a ranking cohort. The two ranking cohorts are not authenticated paired contexts; comparisons between them do not identify ranking improvement.

This establishes response-geometry sensitivity within each saved ranking cohort. It does not establish that Q-IG and L2-IG select the same features, that their rankings are equivalent, or that either is effective. The attempted cross-cohort join has different reference-action norms at every nominal key; exact raw contexts and noise are unavailable. Apparent agreement after common rescoring therefore cannot replace a prospective paired ranking comparison. Likewise, the larger native L2 score is not by itself recovered information, and stabilized L2 has no square-root kink at exact agreement.

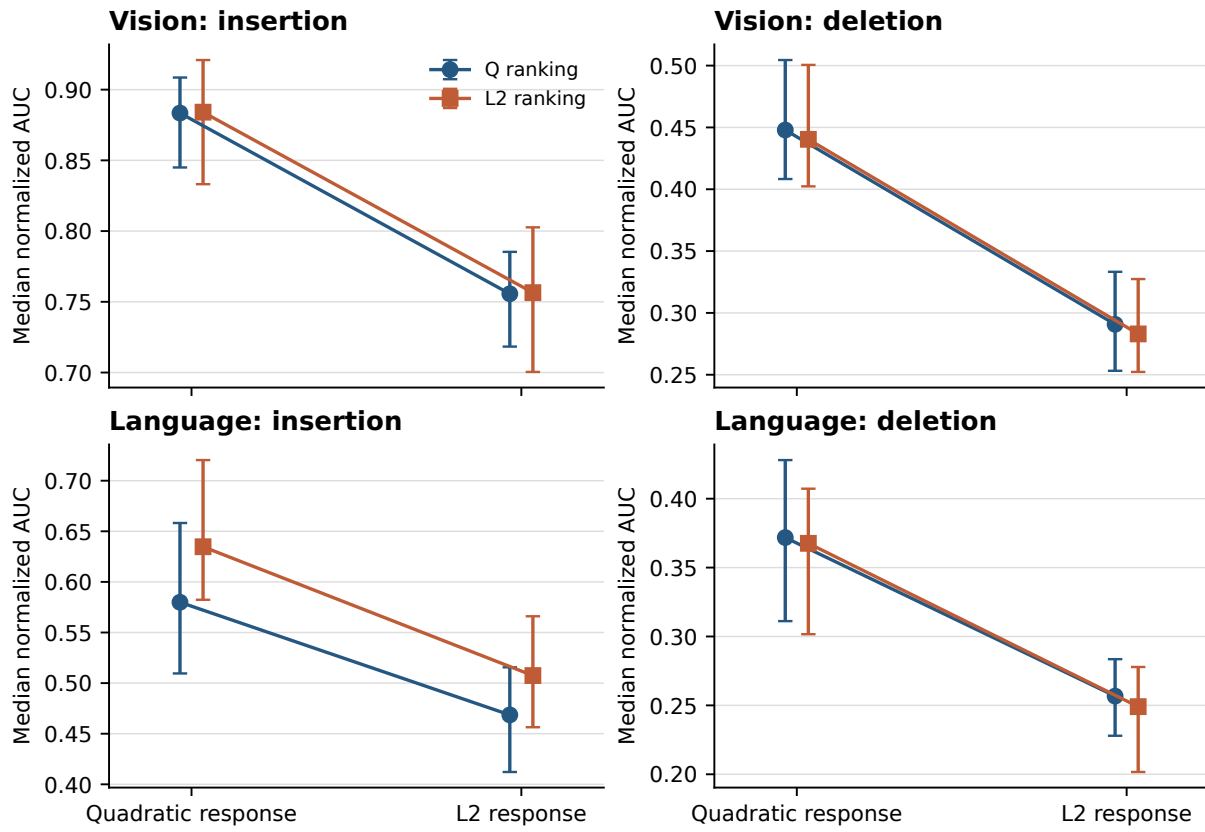


Figure 1: Changing only the evaluation response while keeping each recorded ranking and its interventions fixed. Error bars are conditional episode-bootstrap intervals for each median. Lines connect responses within a ranking cohort; the two ranking cohorts are not authenticated pairs. There is no universal chance reference line.

5.2 An Analytic Random-Order Counterexample

For an additive scalar policy with equal feature contributions, an all-one input and a zero baseline, every feature ordering is identical for perturbation evaluation. At deleted fraction q , quadratic-response normalization gives $1 - q^2$; insertion gives $1 - (1 - q)^2$. Both continuous AUCs are $2/3$, despite the absence of ranking information. Trapezoidal integration on the historical requested grid gives 0.65976 for both curves. The continuous integral and sampled-grid statistic are distinct quantities. An unstabilized negative-norm response instead produces linear normalized curves in this example. Consequently, a fixed AUC reference such as 0.5 requires justification for the policy, response and intervention protocol; it is not universal. This construction is an analytic control, not an empirical random-ranking measurement on RDT.

5.3 Completeness and Perturbation Summaries

Table 2 reports completeness residuals from the saved verification campaign. Vision and language have smaller median relative residuals than state in these cohorts, but this does not establish more accurate coordinates or a causal explanation based on dimension count. The historical integrator combines model-forward, path, summation and quadrature errors. Actual-model validation must separate those factors on identical inputs before selecting production settings.

Task	Calls	Episodes	Vision (%)	Language (%)	State (%)
PegInsertion	700	100	2.17 [1.94, 2.35]	1.83 [1.69, 1.98]	3.48 [3.00, 4.04]
PickCube	397	100	2.20 [1.90, 2.43]	1.88 [1.62, 2.30]	5.88 [4.99, 7.29]
PickYCB	397	100	1.85 [1.71, 2.06]	2.02 [1.79, 2.18]	4.26 [3.50, 5.62]
StackCube	400	100	2.39 [2.18, 2.59]	1.76 [1.58, 1.97]	10.53 [8.07, 13.13]

Table 2: Month 7 verification campaign, all valid calls, $m = 64$. Cells are median relative completeness residuals in percent with conditional 95% episode-bootstrap intervals. Calls and episodes are the identical point/interval populations. These are historical numerical diagnostics, not coordinate-accuracy certifications.

Table 3 reports native quadratic-response insertion and deletion AUCs for the same verification campaign. These are descriptive perturbation results. Without matched random and simple ranking controls, their magnitudes do not establish attribution efficacy. Normalization can produce extreme negative values: the smallest saved PickCube vision insertion AUC in the all-valid population is -18190.36. The analysis retains that observation and its endpoint denominator instead of clipping it or treating the action-norm filter as a validity certificate. Medians are consequently accompanied by the released denominator/tail diagnostics; a central summary alone does not describe the entire curve population.

Task	Modality	Calls	Episodes	Insertion AUC	Deletion AUC
PegInsertion	Vision	700	100	0.731 [0.688, 0.774]	0.280 [0.234, 0.309]
PegInsertion	Language	700	100	0.320 [0.257, 0.379]	0.341 [0.291, 0.382]
PickCube	Vision	397	100	0.822 [0.771, 0.838]	0.167 [0.137, 0.213]
PickCube	Language	397	100	0.525 [0.468, 0.580]	0.433 [0.391, 0.475]
PickYCB	Vision	397	100	0.844 [0.808, 0.863]	0.337 [0.296, 0.372]
PickYCB	Language	397	100	0.335 [0.227, 0.409]	0.471 [0.438, 0.500]
StackCube	Vision	400	100	0.821 [0.777, 0.849]	0.208 [0.161, 0.261]
StackCube	Language	400	100	0.359 [0.292, 0.432]	0.480 [0.429, 0.512]

Table 3: Month 7 verification campaign, all valid calls and the native quadratic response. Medians and conditional 95% intervals share one population. No entry is labeled a success against an uncalibrated chance threshold. The recorded token-fraction grids, endpoint conventions and provenance limits remain part of these descriptive estimates.

6 Numerical Diagnostics from an Incomplete Campaign

The new diagnostic campaign concerns identified current computations, separately from the historical records above. It was stopped for compute cost before its planned numerical coverage and independent verification were complete. The analysis uses a fixed, authenticated snapshot of sealed results; unfinished entries are not classified as successful or failed calculations. This stopping decision provides no statistical sampling rule for the completed subset, so its criterion counts are descriptive rather than estimates of population failure rates.

Functions and precision controls. Authenticated stored observations, masks, reference actions and exact noise arrays permit comparisons on fixed inputs. The diagnostic implementation separates accumulation arithmetic, master-path and final-product arithmetic, quadrature and downstream model precision. It compares full coordinate arrays, eligible-group ranks, selected groups and perturbation responses. An earlier downstream-fp32 factor control lifts the same bf16-rounded parameters and cached adapted representations, explicitly converts non-parameter dtype attributes, and holds the original reference action fixed. That comparison changes downstream arithmetic while holding the numeric inputs fixed; it does not recover unrounded checkpoint parameters or test fp32 encoders. Its fixed original reference also distinguishes it from the subsequent probe. The earlier comparisons preceded the full repeated-gradient checks, so they do not alone establish a globally repeatable precision contrast.

The later engineering campaign uses a separately specified deterministic fp32 function: image and language representations retain their cached bf16 values lifted to fp32, raw state is re-adapted in fp32, and the function generates its own fp32 action reference. It evaluates contexts visited by the bf16 policy with the stored source noise. This is an offline intervention function, not the bf16 behavior policy or a derivative of its executed commands. Repeating its endpoints tests a different property from repeating its path gradients and attribution maps. Changing the function or its reference cannot be treated as a quadrature-only comparison.

Engineering design and interpretation. The predeclared engineering bank spans four RDT-170M tasks and two tasks with an independently identified authors’ 1B checkpoint. Its identity does not recover or authenticate the checkpoint used by the historical task-finetuned 1B cohort in Appendix B. Source collection stopped after at most 13 policy calls, subject also to a 400-environment-step cap and earlier termination; call indices 0 and 12 were selected in advance. These are fixed-index engineering contexts, not uniformly sampled calls from complete trajectories. Collection did not select episodes by success or action norm. The integration ladders use trapezoidal quadrature with fp32 path and accumulation arithmetic. The same gradient stream supplies a mean path-gradient map without the IG input-difference multiplier; it is included in the numerical checks, not evaluated as an efficacy baseline. The numerical protocol compares coordinates, ranks, completeness and common perturbation responses across budgets and repeated calculations. Its tolerance checks diagnose the tested approximation; they are not efficacy margins or calibrated bounds on the error against an exact attribution. The finest available map is a comparator, not ground truth. The preserved cohort analysis reports every planned context, including calls not reached by the source rollout and entries without a sealed result.

Preserved coverage. The retained snapshot is labeled October 1, 2026, 03:36 UTC, before the later compute stop. Table 4 retains all 24 planned contexts and 144 target/modality cells. Twelve contexts, representing six episodes from the four 170M tasks, have all six diagnostics complete, giving 72 complete cells. Three planned calls were not reached by their source rollouts, accounting for 18 cells. The remaining nine available contexts, or 54 cells, lack sealed diagnostics at this cutoff. There is no completed 1B numerical context in this snapshot. These are snapshot statuses, not assertions about the final state of later interrupted work.

Repeatability and budget sensitivity. The image and language ladders use $m = 128,256,512$; the state ladder uses $m = 256,512,1024$. Finest maps were repeated twice, and seven fixed path locations were probed three times each. All recorded repeated-condition equality and coverage checks satisfied their conditions. They are dependent checks on the same finite contexts, not independent replications or evidence of a population reliability rate.

Stratum	Planned	Complete	Unreached	Unsealed	Cells
PegInsertionSide 170M	4	2	0	2	12/24
PickCube 170M	4	4	0	0	24/24
PickCube 1B	4	0	2	2	0/24
PickSingleYCB 170M	4	2	0	2	12/24
StackCube 170M	4	4	0	0	24/24
StackCube 1B	4	0	1	3	0/24
Total	24	12	3	9	72/144

Table 4: Full planned engineering roster at the authenticated 03:36 UTC snapshot. The first four numerical columns count contexts; the last gives complete/planned target-modality cells. Each context has six cells. Unreached source calls and available but unsealed results remain distinct. Completion does not imply numerical qualification.

Table 5 reports every completed target/modality group using the two finest recorded budgets. For maps A_m and A_{2m} , the coordinate discrepancy is $\|A_m - A_{2m}\|_1 / \|A_{2m}\|_1$, undefined at a zero denominator. The predeclared relative-coordinate and completeness thresholds are each 1%; additional checks concern group ordering and common-response sensitivity. The table’s violation counts refer to context/target/modality arms with at least one pairwise-threshold violation in this finer-budget comparison, including the mean path-gradient diagnostics; finest-budget completeness is summarized separately. All lower-budget comparisons, raw denominators and individual check results remain in the artifact. The maxima summarize the recorded cases without an interval or population generalization.

Modality	Target	n	Viol.	IG L_1 (%)	Residual (%)	RMS (%)
Language	L	12	0	0.402	0.315	0.000
Language	Q	12	0	0.006	0.002	0.000
State	L	12	2	20.778	6.655	18.573
State	Q	12	2	32.119	11.436	1.641
Vision	L	12	4	1.234	0.288	7.361
Vision	Q	12	1	2.049	0.124	0.000

Table 5: Diagnostics on all 12 completed contexts per target/modality arm. n counts contexts, and “Viol.” counts arms with at least one recorded pairwise-threshold violation between the two finest budgets. The remaining columns are maxima across those contexts: relative IG coordinate L_1 difference, finest-budget relative completeness residual, and maximum raw RMS response-curve change divided by the all-baseline RMS displacement, all in percent. RMS maxima include IG and mean-path-gradient rankings and both perturbation directions. Coordinate differences use the finer map’s L_1 norm. Comparators are 256 versus 512 for vision/language and 512 versus 1024 for state. These dependent engineering diagnostics are not population error rates or proof that the finest maps are accurate; no setting or subset is approved.

Both targets retain finest-budget state completeness violations at the two recorded PickSingleYCB episode-1 contexts, alongside substantial changes between the two finest maps. Their scalar gaps exceed the declared near-zero cutoffs; the corresponding signed gaps and absolute residuals are retained in the artifact. These discrepancies therefore cannot be dismissed as division by an almost-zero scalar gap. A state case also has unchanged intervention response curves despite substantial coordinate change, showing why response stability alone is insufficient to qualify the integral. Since the quadratic target also fails, the failures cannot all be assigned to the stabilized negative-norm target’s endpoint behavior. The sparse stored path probes do not identify an exclusive numerical cause.

Repeated equality can coexist with integration-budget sensitivity. A deterministic calculation can reproduce an approximation whose value changes with the integration budget, while a small completeness residual can conceal coordinate errors through cancellation. Accordingly, neither repeatability nor a single completeness statistic qualifies the production setting. The observed failures support a bounded numerical diagnostic

conclusion; the incomplete campaign does not establish accuracy across modalities, tasks, checkpoints or future contexts.

7 Discussion and Limitations

The reconstructible evidence separates three questions: what a scalar response measures, whether a numerical attribution is reliable, and whether its ranking contains useful model-dependent information. The saved response transformation answers the first question for fixed recorded interventions. It cannot answer the latter two by itself. The present claims concern exact relationships and reconstructed measurements, supported by the explicitly bounded numerical diagnostics. A new paired study with common responses and informative controls would be needed to establish ranking efficacy; no such efficacy conclusion is drawn here. The elementary identities are not presented as a new general theory of attribution. Their value here is to identify which operation changes a reported evaluation and to prevent that change from being misread as empirical ranking improvement. A common response itself defines the comparison’s estimand, rather than a universal measure of explanation quality. Its action coordinates, units, baselines, perturbation groups and aggregation require justification for the intended use. For the unresolved ranking question, matched random, input-difference and path-gradient controls would help distinguish information in the attribution from simple input structure and the IG multiplier. No comparative or learned-weight-specific result from the deferred protocols is claimed in this paper.

Historical uncertainty also has several distinct sources. Source-record hashing can authenticate released bytes while checkpoint and context identities remain unknown. Episode resampling addresses within-episode dependence conditional on a particular empirical mixture; it does not repair shared or overwritten contexts, preprocessing confounds, or missing training replicates. The 1B results come from one checkpoint, and the solver and alternative-target analyses cover narrower task populations than the verification campaign. Normalized response curves may be unbounded and sensitive to endpoint gaps. Camera-token and language-token interventions differ in group counts and semantics, so their AUCs are not directly interchangeable measures of modality importance.

Historical overlay panels are preserved as artifacts but are not used as evidence of successful grounding or behavioral failure diagnosis. Likewise, groups previously named from completeness residuals are numerical diagnostic groups, not independently validated labels such as misgrounding or delayed credit. The unrecovered original main-pass and early budget-extension numbers are excluded from revised primary tables. Appendix D reports only reconstructible later budget cohorts, without a convergence claim.

8 Conclusion

Per-policy-call IG for a fixed-noise diffusion sampler can be defined using an explicit auxiliary action-discrepancy score. For saved RDT perturbation curves, changing only the evaluation response changes normalized AUC without changing the selected features. An analytic example rules out a universal random-ranking AUC of 0.5. The canonical retrospective tables preserve what the available records support and exclude unrecovered primary constants. The actual-model diagnostics retain unresolved numerical failures and incomplete coverage. A common response is necessary for an interpretable comparison of ranking quality, but our results do not establish that one target or attribution method is superior under it.

Ethical Statement

This work analyzes pretrained robotic manipulation policies in simulation; no human subjects were involved. Attribution is a diagnostic tool here, not a safety guarantee for robot deployment. Misinterpreting attractive maps, low completeness residuals or uncalibrated AUCs could create unjustified confidence in a policy, which motivates the explicit limits and controls.

Reproducibility Statement

The retrospective analysis parses a hash-authenticated raw inventory, records occurrence decisions, and links each result to physical source records. The revised tables and figures are generated from the same canonical artifact, with identical point/interval populations and a manuscript lineage registry. It retains conflicting-key exclusions, action-norm strata, endpoint/tail diagnostics, alternative episode weightings and conditional grouping sensitivities. These operations reproduce arithmetic over saved records; they do not recover missing models, sidecars or original observations. The portable numerical projection retains the full planned roster, recorded checks and metrics, report hashes, and selected YCB FP32 tensors with their tensor hashes; it does not include all raw artifacts for the 72 completed target-modality cells. Its CPU reproducer recomputes the reported diagnostics and saved-tensor checks from this projection, not the complete original raw-artifact audit or a new model execution. The new runtime separately records immutable context, source, checkpoint, language, environment and noise identities, and fails explicitly on nonfinite attribution rather than sanitizing coordinates. A candidate PDF or a successful script run alone does not constitute an authenticated scientific replication. The identified repository URL is omitted from this draft. An anonymous review supplement must be assembled and inspected after the scientific gates are complete.

References

- Julius Adebayo, Justin Gilmer, Michael Muelly, Ian J. Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, pp. 9525–9536, 2018. URL <https://papers.nips.cc/paper/2018/hash/294a8ed24b1ad22ec2e7efea049b8737-Abstract.html>.
- Yotam Amitai, Yael Septon, and Ofra Amir. Explaining reinforcement learning agents through counterfactual action outcomes. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 10003–10011, 2024. URL <https://ojs.aaai.org/index.php/AAAI/article/download/28863/29640>.
- Mariusz Bojarski, Anna Choromanska, Krzysztof Choromanski, Bernhard Firner, Larry Jackel, Urs Muller, P. Yeres, and Karol Zieba. VisualBackProp: Efficient visualization of CNNs for autonomous driving. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4701–4708, 2018. doi: 10.1109/ICRA.2018.8461053. URL <https://doi.org/10.1109/ICRA.2018.8461053>.
- Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023. URL <https://roboticsproceedings.org/rss19/p026.html>.
- Samuel Greydanus, Anurag Koul, Jonathan Dodge, and Alan Fern. Visualizing and understanding Atari agents. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pp. 1792–1801, 2018. URL <https://proceedings.mlr.press/v80/greydanus18a.html>.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 6840–6851, 2020. URL <https://papers.nips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>.
- Sara Hooker, Dumitru Erhan, Pieter-Jan Kindermans, and Been Kim. A benchmark for interpretability methods in deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, pp. 9734–9745, 2019. URL https://papers.nips.cc/paper_files/paper/2019/hash/fe4b8556000d0f0cae99daa5c5c5a410-Abstract.html.
- Sumit Kumar Jha, Rickard Ewetz, Alvaro Velasquez, and Susmit Jha. On smoother attributions using neural stochastic differential equations. In *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 522–528, 2021. URL <https://www.ijcai.org/proceedings/2021/73>.
- Sumit Kumar Jha, Rickard Ewetz, Alvaro Velasquez, Arvind Ramanathan, and Susmit Jha. Shaping noise for robust attributions in neural stochastic differential equations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pp. 9567–9574, 2022. URL <https://ojs.aaai.org/index.php/AAAI/article/view/21190>.

- Andrei Kapishnikov, Subhashini Venugopalan, Besim Avci, Ben Wedin, Michael Terry, and Tolga Bolukbasi. Guided integrated gradients: An adaptive path method for removing noise. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5050–5058, 2021. URL https://openaccess.thecvf.com/content/CVPR2021/html/Kapishnikov_Guided_Integrated_Gradients_An_Adaptive_Path_Method_for_Removing_Noise_CVPR_2021_paper.html.
- Jinwoong Kim and Sangjin Park. IMPACT-VLA: Interaction-aware multimodal propagation attribution via counterfactual trajectories for vision-language-action policies. *arXiv preprint arXiv:2609.15005*, 2026. URL <https://arxiv.org/abs/2609.15005>.
- Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. RDT-1B: a diffusion foundation model for bimanual manipulation. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025. URL <https://arxiv.org/html/2410.07864v2>.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models. *Machine Intelligence Research*, 22(4):730–751, 2025. URL <https://mi-research.net/article/doi/10.1007/s11633-025-1562-4>.
- Octo Model Team, Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, et al. Octo: An open-source generalist robot policy. In *Proceedings of Robotics: Science and Systems (RSS)*, 2024. URL <https://www.roboticsproceedings.org/rss20/p090.pdf>.
- Vitali Petsiuk, Abir Das, and Kate Saenko. RISE: Randomized input sampling for explanation of black-box models. In *Proceedings of the British Machine Vision Conference (BMVC)*, 2018. URL <https://www.bmvc2018.org/contents/papers/1064.pdf>.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020. URL <https://jmlr.org/papers/v21/20-074.html>.
- Yao Rong, Tobias Leemann, Vadim Borisov, Gjergji Kasneci, and Enkelejda Kasneci. A consistent and efficient evaluation strategy for attribution methods. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pp. 18770–18795, 2022. URL <https://proceedings.mlr.press/v162/rong22a.html>.
- Haoyuan Shi, Xiancong Ren, Yingji Zhang, Qinfan Zhang, Jiayu Hu, Haozhe Shan, Han Dong, Jinpeng Lu, Yinda Chen, Yi Zhang, Yong Dai, and Xiaozhu Ju. VLA-Trace: Diagnosing vision-language-action models through representation and behavior tracing. *arXiv preprint arXiv:2605.30117*, 2026. URL <https://arxiv.org/abs/2605.30117>.
- Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pp. 3319–3328, 2017. URL <https://proceedings.mlr.press/v70/sundararajan17a.html>.
- Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus Stenetorp, Jimmy Lin, and Ferhan Ture. What the DAAM: Interpreting Stable Diffusion using cross attention. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 5644–5659, 2023. URL <https://aclanthology.org/2023.acl-long.310/>.
- Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, Yulin Liu, Tse-kai Chan, et al. ManiSkill3: GPU parallelized robotics simulation and rendering for generalizable embodied AI. *arXiv preprint arXiv:2410.00425*, 2024. URL <https://arxiv.org/abs/2410.00425>.
- Richard Tomsett, Dan Harborne, Supriyo Chakraborty, Prudhvi Gurram, and Alun D. Preece. Sanity checks for saliency metrics. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pp. 6021–6029, 2020. URL <https://ojs.aaai.org/index.php/AAAI/article/view/6064>.

- Chase Walker, Sumit Kumar Jha, Kenny Chen, and Rickard Ewetz. Integrated decision gradients: Compute your attributions where the model makes its decision. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 5289–5297, 2024. URL <https://ojs.aaai.org/index.php/AAAI/article/view/28336>.
- Chase Walker, Md Rubel Ahmed, Sumit Kumar Jha, and Rickard Ewetz. Explaining ViTs using information flow. In *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, pp. 2440–2448, 2025a. URL <https://proceedings.mlr.press/v258/walker25a.html>.
- Chase Walker, Sumit Kumar Jha, and Rickard Ewetz. Metric-driven attributions for vision transformers. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025b. URL <https://openreview.net/pdf?id=rGP2jBWt0l>.
- Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 11975–11986, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Zhai_Sigmoid_Loss_for_Language_Image_Pre-Training_ICCV_2023_paper.html.
- Hanxin Zhang, Mingshuo Xu, Abdulqader Dhafer, Shigang Yue, Hongbiao Dong, and Zhou Daniel Hao. Embodied interpretability: Linking causal understanding to generalization in vision-language-action models. *arXiv preprint arXiv:2605.00321*, 2026. URL <https://arxiv.org/abs/2605.00321>.
- Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. In *Proceedings of the 7th Conference on Robot Learning (CoRL)*, pp. 2165–2183, 2023. URL <https://proceedings.mlr.press/v229/zitkovich23a.html>.

A Response-Geometry Derivations

The identities in Section 3.4 assume that the relevant path gradients are integrable and hold the action reference fixed. Differentiating $Q = -r/(2D\sigma^2)$ and $L = -\sqrt{r+\epsilon}$ gives Equation 5 immediately. For Equation 11, set $a_0 = \sqrt{\epsilon}$, $b = \sqrt{R+\epsilon}$ and $z = \sqrt{r+\epsilon}$. Since $R = b^2 - a_0^2$ and $r = z^2 - a_0^2$, rationalization gives

$$\frac{b^2 - z^2}{R} - \frac{b - z}{b - a_0} = \frac{(z - a_0)(b - z)}{R}. \quad (12)$$

For in-range residuals the maximum difference is $(b - a_0)^2/(4R) \leq 1/4$. These pointwise statements concern a fixed ranking’s interventions; they neither order different rankings nor establish task-success improvements.

For the rank-reversal example, let $c = 4/5$ and parameterize the path by (t, t) . The scalar residual is $d(t) = 1 + c - t^2 - ct = (1 - t)(1 + t + c) \geq 0$. The quadratic path integrands are $2td(t)$ and $cd(t)$, yielding

$$\text{IG}^Q = \left(\frac{1}{2} + \frac{c}{3}, \frac{2c}{3} + \frac{c^2}{2} \right) = \left(\frac{23}{30}, \frac{64}{75} \right). \quad (13)$$

For the stabilized negative norm the respective integrands are $2td(t)/\sqrt{d(t)^2 + \epsilon}$ and $cd(t)/\sqrt{d(t)^2 + \epsilon}$. Using $d(t) \geq c(1 - t)$ and monotonicity of $1 - d/\sqrt{d^2 + \epsilon}$ gives

$$\int_0^1 \left(1 - \frac{d(t)}{\sqrt{d(t)^2 + \epsilon}} \right) dt \leq 1 - \frac{\sqrt{c^2 + \epsilon} - \sqrt{\epsilon}}{c} \leq \frac{\sqrt{\epsilon}}{c}. \quad (14)$$

Therefore $0 \leq 1 - \text{IG}_1^L \leq 2\sqrt{\epsilon}/c = 2.5 \times 10^{-6}$ and $0 \leq c - \text{IG}_2^L \leq \sqrt{\epsilon} = 10^{-6}$ at the implemented stabilizer. All coordinates are positive and the stabilized L margin is at least 0.1999975, while Q has the opposite ordering with exact margin 13/150. The artifact includes independent differentiation, integration and normalization checks for these identities, including overshoots, zero gaps and a policy example without rank reversal.

B Secondary Historical Results

These analyses preserve additional observations from the saved campaigns. Their training, reference, context and preprocessing qualifications prevent interpreting them as model-scale effects, learned-weight-specific efficacy or a causal test of denoiser contraction.

B.1 The Separately Recorded 1B Cohort

The saved 1B analysis illustrates a second distinction: changing the episode weighting changes the estimand. Table 6 uses the same 1,199 calls from 60 episodes under both summaries. The language summaries are more sensitive to this weighting than the vision summaries, within a cohort that also mixes language-baseline provenance. Neither estimate is a training-replication result or a controlled improvement over 170M.

Estimand	Modality	Insertion AUC	Deletion AUC
Median over calls	Vision	0.940 [0.926, 0.954]	0.288 [0.250, 0.312]
Median over calls	Language	0.541 [0.449, 0.605]	0.267 [0.187, 0.327]
Median of episode medians	Vision	0.940 [0.927, 0.959]	0.289 [0.235, 0.313]
Median of episode medians	Language	0.488 [0.368, 0.581]	0.297 [0.151, 0.342]

Table 6: Saved RDT-1B PickCube campaign, 1,199 source calls and 60 episodes under both estimands. The first weights each call equally; the second weights each episode’s median equally. Conditional 95% episode intervals describe the released mixture. Evaluation seeds share a checkpoint, and the pooled language population spans two baseline provenances.

B.2 Randomization and Baseline Diagnostics

Table 7 reconstructs the standard verification-campaign correlations. Model randomization retains substantial rank association, while input shuffling changes the maps more strongly. That contrast does not make model dependence unnecessary or certify semantic grounding. A map based only on input magnitude can change under input shuffling while ignoring learned policy weights. The saved frozen-reference and cascade variants, including a dependence-grouping sensitivity, are reported in Appendix C.

Task	Check	Calls	Episodes	Vision ρ	Language ρ
PegInsertion	C1	100	16	0.960 [0.949, 0.969]	0.959 [0.947, 0.966]
PegInsertion	C2	100	16	0.089 [0.067, 0.111]	0.018 [0.004, 0.036]
PickCube	C1	100	26	0.956 [0.948, 0.961]	0.926 [0.917, 0.944]
PickCube	C2	100	26	0.142 [0.128, 0.157]	0.117 [0.099, 0.150]
PickYCB	C1	100	26	0.943 [0.930, 0.949]	0.917 [0.903, 0.931]
PickYCB	C2	100	26	0.164 [0.149, 0.175]	0.128 [0.081, 0.155]
StackCube	C1	100	26	0.958 [0.952, 0.965]	0.965 [0.960, 0.969]
StackCube	C2	100	26	0.140 [0.117, 0.154]	0.266 [0.246, 0.279]

Table 7: Month 7 model randomization (C1) and input shuffle (C2), all recorded calls. Conditional 95% episode intervals accompany median Spearman correlation. These diagnostics test different dependencies: low C2 correlation does not establish learned-model dependence, and C1 changes integration budget as well as weights.

The IG multiplier ($x_i - x'_i$) permits shared structure but does not force a particular ranking. The path gradients can dominate the result, and the input features themselves come from retained learned encoders/adaptors. Persistent correlations therefore motivate input-only and path-gradient-only controls; they do not establish multiplier dominance. Spearman correlation is not a fraction of model-independent information. The baseline comparisons in Appendix C show imperfect agreement but cannot establish equivalence with their small number of groups.

B.3 Action Displacement and Solver Resolution

The quadratic score is a normalized squared displacement, so its numerical magnitude has a direct dependence on the chosen convention. Removing $1/D$ multiplies exact scores and attributions by D without changing exact ranks, normalized AUC, or relative completeness. This algebra does not determine whether an action change is operationally important; that requires action units and a justified behavioral criterion.

The saved solver analysis compares median relative action displacement under top-5% vision deletion between $T = 2$ and $T = 20$. Figure 2 reports the endpoint contrast with conditional resampling of nominal episode pairs. Reference actions vary across solver settings, and exact contexts/noise cannot be authenticated. The intervals permit appreciable increases and decreases. Changing T changes discretization over the full denoising horizon, rather than adding steps along one fixed trajectory. These observations therefore identify neither a causal contraction mechanism nor its absence.

C Additional Retrospective Diagnostics

The frozen-reference and backbone-cascade analyses retain learned encoders/adaptors and historical context uncertainty. Table 8 reports both recorded-episode grouping and a conditional shared-context grouping. These are sensitivity analyses, not proof that either grouping is independent. Table 9 reports baseline-ranking point summaries; the small group count prevents an equivalence conclusion.

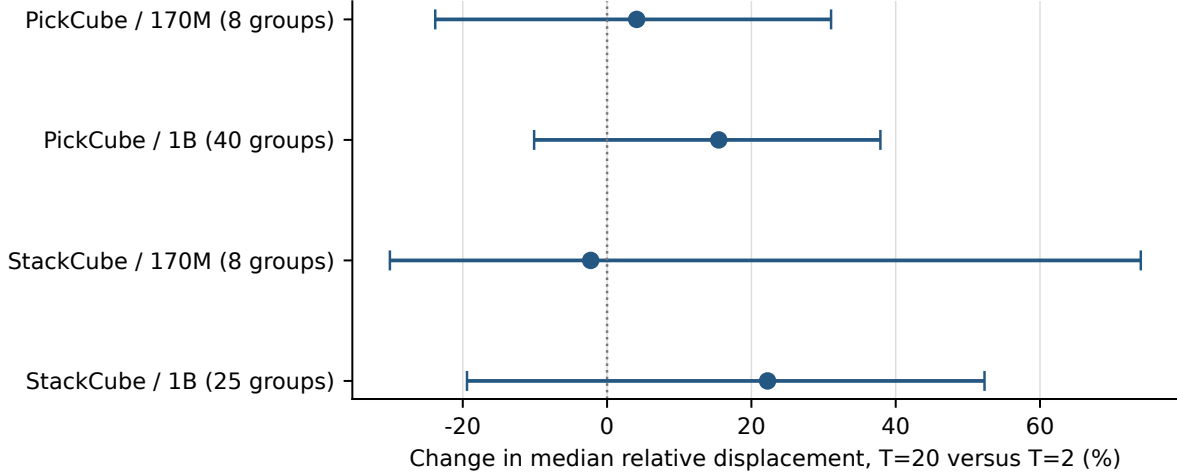


Figure 2: Retrospective change in median relative displacement under top-5% vision deletion. Intervals use nominal episode pairing across solver resolutions, conditional on the released records and grouping. The vertical reference denotes no endpoint change, not a contraction test or an equivalence margin. Raw context/noise identity remains unknown.

Variant	Grouping	Calls	Groups	Vision ρ	Language ρ
Frozen	Recorded episodes	400	16	0.929 [0.905, 0.940]	0.862 [0.835, 0.879]
Frozen	Shared-context sensitivity	400	8	0.929 [0.904, 0.947]	0.862 [0.820, 0.886]
Cascade	Recorded episodes	400	16	0.713 [0.696, 0.738]	0.910 [0.896, 0.920]
Cascade	Shared-context sensitivity	400	8	0.713 [0.693, 0.763]	0.910 [0.894, 0.926]

Table 8: Saved backbone-randomization variants. Shared-context grouping joins nominal task/episode keys across recorded seeds as a dependence sensitivity; it does not authenticate shared inputs. Intervals are conditional on the stated grouping. Learned encoders/adaptors remain, and sidecar reuse and language-baseline deviations remain unresolved.

Task	Calls	Groups	Black/gray	Black/blur	Gray/blur
PickCube	30	2	0.825	0.539	0.455
StackCube	30	2	0.814	0.546	0.496

Table 9: Saved baseline-ranking correlations. Only two nominal episode groups are available per task. Interval output is retained in the analysis artifact as a descriptive resampling diagnostic, but is not presented here as calibrated uncertainty. Positive correlation does not establish baseline interchangeability.

D Saved Integration-Budget Cohorts

Table 10 replaces the previous table of unrecovered and duplicated constants with the reconstructible later cohorts. The norm-selected population and occurrence rule are retained explicitly for this historical sensitivity. These calls are not paired with the earlier $m = 64$ main pass on identical observations, baselines, runtime and noise. Truncating trajectories to the same call depth does not create such pairs. An integration-budget effect must instead be evaluated on authenticated fixed contexts, with precision and quadrature controlled.

Task	Seed	Calls	Groups	Vision residual $\leq 3\%$ (%)
PegInsertion	42	189	8	64.0 [50.3, 76.9]
PegInsertion	142	160	7	76.2 [67.1, 85.0]
PickYCB	42	194	8	73.7 [62.1, 85.5]
PickYCB	142	195	8	71.8 [60.9, 83.1]
PegInsertion (conflicts excluded)	142	142	7	75.4 [63.8, 85.0]

Table 10: Reconstructible $m = 128$ cohorts only. These conditional intervals describe recorded cohorts, not convergence under paired contexts. The final row removes every conflicting repeated key. Missing original Pick-Cube/StackCube subsets are not reconstructed or replaced with constants.